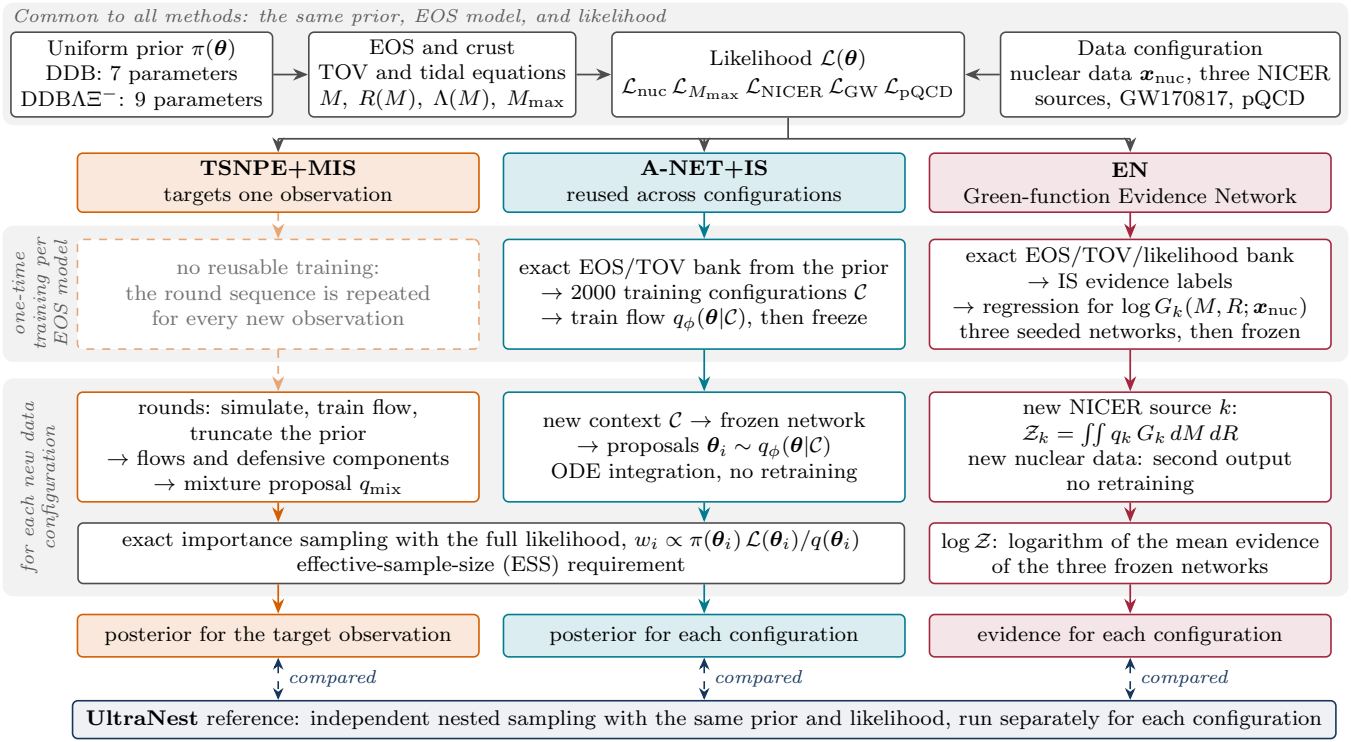
Common to all methods: the same prior, EOS model, and likelihood
Uniform prior $\pi(\boldsymbol{\theta})$
DDB: 7 parameters
DDBΛΞ⁻: 9 parameters
EOS and crust
TOV and tidal equations
M, $R(M)$, $\Lambda(M)$, $M_{\max}$
Likelihood $\mathcal{L}(\boldsymbol{\theta})$
$\mathcal{L}_{\rm nuc}\,\mathcal{L}_{M_{\max}}\,\mathcal{L}_{\rm NICER}\,\mathcal{L}_{\rm GW}\,\mathcal{L}_{\rm pQCD}$
Data configuration
nuclear data $\boldsymbol{x}_{\rm nuc}$, three NICER sources, GW170817, pQCD
TSNPE+MIS
targets one observation
A-NET+IS
reused across configurations
EN
Green-function Evidence Network
one-time training per EOS model
no reusable training:
the round sequence is repeated
for every new observation
exact EOS/TOV bank from the prior
→ 2000 training configurations $\mathcal{C}$
→ train flow $q_\phi(\boldsymbol{\theta}|\mathcal{C})$, then freeze
exact EOS/TOV/likelihood bank
→ IS evidence labels
→ regression for $\log G_k(M, R; \boldsymbol{x}_{\rm nuc})$
three seeded networks, then frozen
for each new data configuration
rounds: simulate, train flow,
truncate the prior
→ flows and defensive components
→ mixture proposal $q_{\rm mix}$
new context $\mathcal{C}$ → frozen network
→ proposals $\boldsymbol{\theta}_i \sim q_\phi(\boldsymbol{\theta}|\mathcal{C})$
ODE integration, no retraining
new NICER source k:
$\mathcal{Z}_k = \iint q_k\, G_k\, dM\, dR$
new nuclear data: second output
no retraining
exact importance sampling with the full likelihood, $w_i \propto \pi(\boldsymbol{\theta}_i)\,\mathcal{L}(\boldsymbol{\theta}_i)/q(\boldsymbol{\theta}_i)$
effective-sample-size (ESS) requirement
$\log \mathcal{Z}$: logarithm of the mean evidence
of the three frozen networks
posterior for the target observation
posterior for each configuration
evidence for each configuration
compared
compared
compared
UltraNest reference: independent nested sampling with the same prior and likelihood, run separately for each configuration